

WHY We Are Doing This Research

Scientific motivation, research gap, and value of the experiment

Research Project: Cross-Cycle Explanation Drift in Solar Flare Forecasting

Domain: Space Weather | Machine Learning | Explainable AI

Status: Locked research direction - October 2026

Purpose

Use this document to explain why prediction accuracy alone is not enough, why Solar Cycle 25 is a meaningful transfer test, and why explanation stability matters.

Based on the locked Master Research Reference v1.0

1. Why This Research Exists

Main reason

Ek ML model ka prediction score acceptable reh sakta hai, lekin model ke feature-attribution patterns change ho sakte hain. Agar hum sirf TSS dekhien, toh ye hidden behavior change miss ho sakta hai.

2. Solar Cycle Change Is a Real Stress Test

Solar activity roughly 11-year cycles mein change hoti hai. Cycle 24 aur Cycle 25 identical data regimes nahi hain. Isliye Cycle 24 par trained model ko Cycle 25 par unchanged test karna ek natural **distribution-shift** / **generalization test**

- Feature distributions change ho sakti hain.
- Flare prevalence change ho sakti hai.
- Active-region population properties change ho sakti hain.
- Model ka performance ya explanation structure in shifts ke under behave differently kar sakta hai.

3. Prediction Score Alone Kyon Enough Nahi Hai?

Simple example

Suppose Cycle 24 TSS = 0.72 aur Cycle 25 TSS = 0.70. Sirf performance dekhkar hum bol sakte hain "model stable hai". Lekin agar SHAP ranking/attribution pattern strongly change ho gaya, model ka behavior internally different ho sakta hai. Ye difference scientifically investigate karna important hai.

Hum SHAP ko human reasoning ya causal physics nahi bolte. SHAP hume **model attribution behavior** ka structured view deta hai.

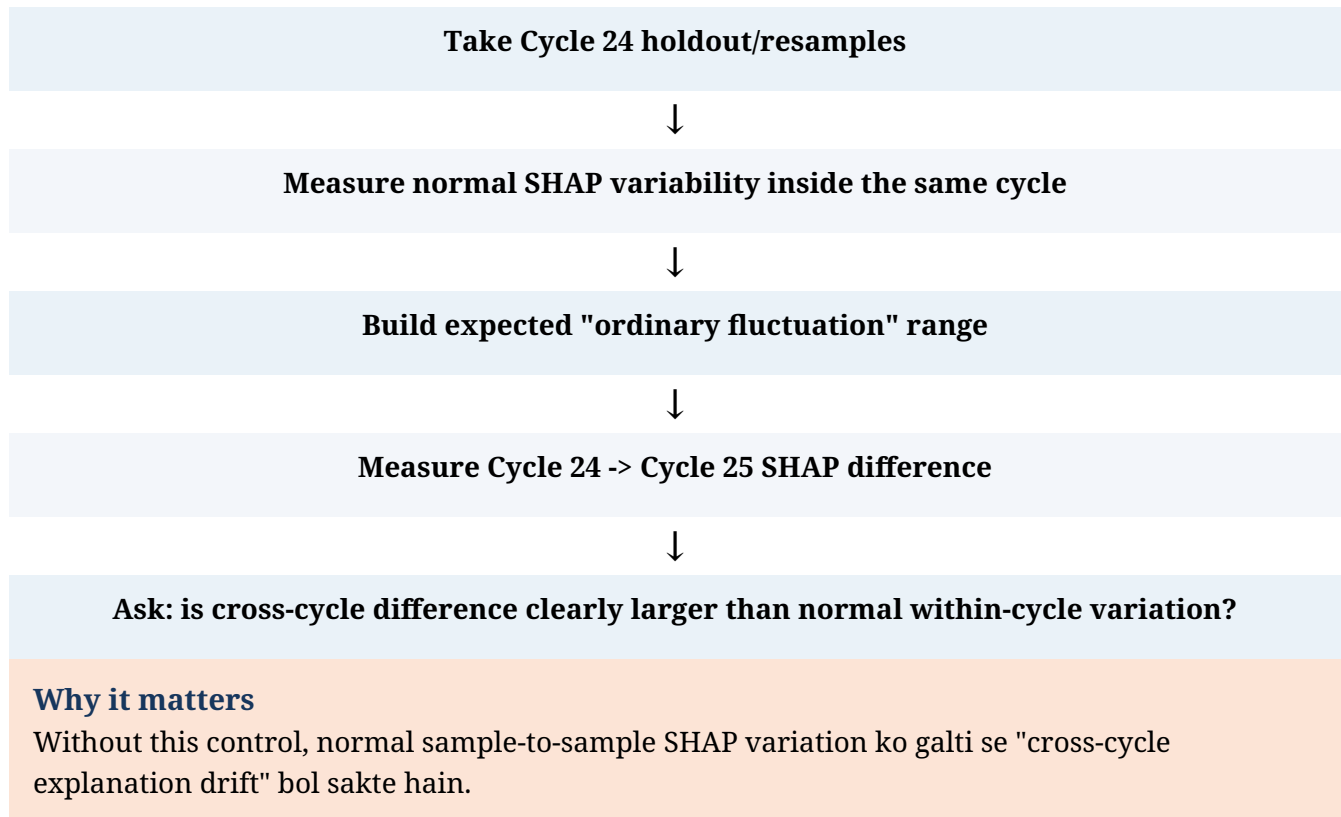
4. Actual Research Gap

Existing literature ne separately ye areas study kiye hain:

- Cross-cycle / temporal solar-flare prediction.
- SHAP or explainable AI for flare forecasting.
- Cycle 25 datasets and benchmarks.

Our narrower gap: same fixed Cycle-24-trained model ke predictive drift aur SHAP explanation drift ko Cycle 24 holdout vs Cycle 25 par controlled, statistically robust way mein compare karna - including a within-Cycle-24 explanation-variability baseline.

5. Why the Within-Cycle-24 Null Baseline Is Essential



6. Why Distribution Shift Must Be Audited

Concept	Simple meaning
Covariate shift	Feature values/distributions Cycle 25 mein Cycle 24 se alag ho sakte hain.
Label-prevalence shift	$\geq M$ -class flare positive rate cycles/blocks ke beech change ho sakta hai.
Missingness / quality shift	Data availability ya measurement quality change ho sakti hai.
AR population shift	Different types/ranges of active regions evaluation sample mein aa sakte hain.

Isliye explanation drift detect hone par hum immediately "model changed its logic because solar physics changed" nahi bolenge. Pehle data-shift explanations investigate karenge.

7. Why Correlated SHARP Features Matter

Important caution

SHARP parameters strongly correlated ho sakte hain. Do correlated features ranking swap karen toh ye automatically genuine physical or model-behavior change prove nahi karta. Isliye feature groups/clusters aur group-level attribution stability bhi check karenge.

8. Scientific Value of Every Outcome

Result	Why it is useful
Stable prediction + stable SHAP	Evidence that evaluated models transfer with stable attribution structure.
Stable prediction + SHAP drift	Shows performance alone may hide a change in model attribution behavior.
Prediction drop + SHAP drift	Shows both generalization and explanation structure are sensitive to cross-cycle shift.
Prediction drop + stable SHAP	Suggests model uses similar features but predictive reliability changes for other reasons.

9. Why This Can Become a Strong Research Contribution

- It asks a narrow and testable question instead of a broad "AI predicts flares" claim.
- It separates predictive performance from explanation behavior.
- It includes a proper null/control baseline.
- It explicitly audits distribution shift and correlated features.
- It is reproducible with public scientific data and tabular ML models.
- It does not depend on obtaining one exciting result; stable and drifting outcomes are both informative.

One-line WHY

Hum ye research isliye kar rahe hain kyunki Cycle 25 jaise new solar regime mein model ki accuracy dekhna alone enough nahi ho sakta; hume ye bhi test karna hai ki same model ka feature-attribution behavior stable hai ya statistically meaningful way mein drift karta hai.